Arista Access Node HSR-ACC-27 QoS Congestion Event SOP

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5625

Description

This Standard Operating Procedure (SOP) outlines the diagnostic, resolution, and verification procedures for addressing QoS congestion events on the Arista Access Node HSR-ACC-27. The SOP is triggered by the occurrence of a QoS congestion event, specifically microbursts, interface tail drops, and QoS queue starvation, as indicated by the ALARM_5625 alarm code with a severity of Warning. This failure domain is critical on the Access role node due to its responsibility for providing high-speed connectivity to end-user devices. The underlying protocol mechanics contributing to this failure domain involve the operation of the Access Node's QoS policies, which are designed to prioritize traffic based on predefined criteria. However, in the event of a congestion event, the QoS policies may become overwhelmed, leading to microbursts, interface tail drops, and QoS queue starvation. These symptoms can have a cascading effect on the network, causing packet loss, jitter, and other service-affecting issues. Therefore, it is essential to promptly diagnose and resolve QoS congestion events on the Access Node HSR-ACC-27 to ensure the continued delivery of high-quality network services.

Pre-requisite Checks

- Check: Verify that the Access Node HSR-ACC-27 is reachable and accessible for remote hands operations.

- Check: Confirm that the redundant link status is normal and functioning correctly.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Outdated or incompatible QoS software image	Medium
Hardware malfunction or failure	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy all, Expected Output: Display all QoS policies configured on the Access Node, Description: This command is used to display all QoS policies configured on the Access Node, including the policy name, priority, and traffic classification criteria. The output will help identify any misconfigured QoS policies that may be contributing to the congestion event.

Command: show interface all statistics, Expected Output: Display interface statistics, including packet drops and queue drops, Description: This command is used to display interface statistics, including packet drops and queue drops, which can help identify the
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interface(s) experiencing congestion.

Command: show qos queue all, Expected Output: Display QoS queue
statistics, including queue length and packet drop rates, Description:
This command is used to display QoS queue statistics, including queue
length and packet drop rates, which can help identify the QoS queues
experiencing congestion.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: conf t set qos policy ..., Description: This step involves verifying and correcting the QoS policy configuration to ensure it is correctly prioritizing traffic and preventing congestion. The exact command will depend on the specific policy configuration and any corrections required.

Step 2: Step: Update QoS software image to latest version, Command: download ..., Description: This step involves updating the QoS software image to the latest version to ensure that any known issues or bugs are resolved. The exact command will depend on the specific image and version being updated to.

Step 3: Step: Perform hardware sanity checks, Command: show hardware all, Description: This step involves performing hardware sanity checks to ensure that the Access Node hardware is functioning correctly and not contributing to the congestion event. The output will help identify any hardware issues that may need to be addressed.

Post-Check Verification

```
Command: show qos policy all, Expected Output: Verify that QoS policy
configuration is correct, Description: This command is used to verify
that the QoS policy configuration is correct and that the congestion
event has been resolved.

Command: show interface all statistics, Expected Output: Verify that
interface statistics are normal, Description: This command is used to
verify that interface statistics, including packet drops and queue
drops, are normal and that the congestion event has been resolved.

Command: show qos queue all, Expected Output: Verify that QoS queue
statistics are normal, Description: This command is used to verify
that QoS queue statistics, including queue length and packet drop
rates, are normal and that the congestion event has been resolved.
```

Roll-back Procedures

- Step: Revert QoS policy configuration changes, Command: rollback ..., Description: This step involves reverting any QoS policy configuration changes made during the resolution process to ensure that the original configuration is restored.

- Step: Revert QoS software image update, Command: rollback ..., Description: This step involves reverting the QoS software image update to the original version to ensure that any changes made during the resolution process are reversed.

- Step: Revert hardware changes, Command: rollback ..., Description: This step involves reverting any hardware changes made during the resolution process to ensure that the original hardware configuration is restored.